

Assignment 5 Report — ML Security & Abuse Pathways

Part 1 — PGD Evasion Audit

Model	ϵ	FPR (AA)	FPR (CA)	AIR	AIR < 0.80?	Note
LR	0.25	0.569	0.370	1.535	No	FPR rising sharply
LR	0.50	0.791	0.560	1.411	No	-----
LR	1.00	0.978	0.884	1.106	No	-----
LR	2.00	1.000	1.000	1.000	No	100% FPR both groups
GBT	0.25	0.317	0.178	1.782	No	Completely unaffected
GBT	0.50	0.317	0.178	1.782	No	Completely unaffected
GBT	1.00	0.317	0.178	1.782	No	Completely unaffected
GBT	2.00	0.317	0.178	1.782	No	Completely unaffected

Table 1. Clean baseline: LR AIR = 1.961, GBT AIR = 1.782. AIR did not cross 0.80 for either model.

Are the two models equally vulnerable? No — they're at opposite extremes. LR is highly sensitive to PGD: FPR for African-American defendants climbs from 0.281 to 1.000 as ϵ goes from 0 to 2.0, and Caucasian defendants follow a similar path. The AIR shrinks from 1.961 toward 1.0 at $\epsilon = 2.0$ because both groups converge to 100% FPR — the model stops discriminating at all, flagging everyone as high-risk. AIR never falls below 0.80 for LR. GBT, on the other hand, is completely immune: FPR and AIR stay fixed at 0.317, 0.178, and 1.782 at every ϵ value. The finite-difference gradient perturbations don't push samples across any tree leaf boundary, so the predictions don't change. For model selection in high-stakes deployments, this means GBT's structural resistance to this type of attack is a real advantage, but it's not a fairness win — GBT's baseline disparity (AIR = 1.782) is already concerning. LR's vulnerability is more operationally dangerous: a white-box attacker can make LR flag everyone as high-risk, collapsing any meaningful risk differentiation.

Part 2 — Poisoning Loop with Fairness Monitoring

(a) AUC and AIR degradation: The clean-model baseline AIR is already 1.961, above the [0.80, 1.25] fairness zone. The African-American-targeted attack makes things worse: AIR rises from 1.961 to 3.010 at 30% poison rate while AUC drops only 0.34pp (from 0.7345 to 0.7311). The Caucasian-targeted attack has minimal effect — AIR barely moves (stays 1.84–2.04 across all poison rates). This asymmetry exists because there are far more AA high-risk training records to flip (1,151 eligible vs. 348 for Caucasian), so the attack has greater leverage on the AA group.

(b) Stealth zone: Since baseline AIR is already above 1.25, the entire tested range (0%–30%) technically constitutes a stealth zone for the AA-targeted attack: AUC never drops more than 0.43pp while AIR increases from 1.96 to 3.01. The more operationally meaningful way to frame it is: at 5% poison rate (57 flipped labels), AIR has already risen to 1.948 with only -0.08pp AUC change — completely invisible to a performance monitor. At 30%, AIR triples the baseline disparity with AUC still looking healthy.

(c) PSI drift monitor: PSI on every feature column is exactly 0.0 at all poison rates for both attack variants, even at 30% poison rate. This is expected — label-flip poisoning only changes training labels, not input features. A PSI-based drift monitor (threshold < 0.10 per feature) would never trigger an alert for either attack. This confirms a fundamental governance gap: PSI detects covariate shift but is completely blind to

label corruption. The only way to catch this is direct fairness monitoring, such as tracking AIR change between model versions.

Part 3 — Membership Inference Depth

(a) Confidence-gap histograms: Both histograms show heavily overlapping distributions between training members and test non-members. LR's mean confidence for training members (0.580) and test non-members (0.581) are nearly identical. GBT shows a slightly wider separation but still very small. This visual overlap explains why MI AUC is near 0.50 for both models — the meta-classifier can't meaningfully separate members from non-members based on confidence scores.

(b) Does gen gap predict MI AUC? The direction is consistent with the hypothesis: GBT has a larger generalization gap (+0.080) and a slightly higher MI AUC (0.500 vs 0.494 for LR). However, both values are essentially random — neither model leaks meaningful membership information. LR actually has a negative generalization gap (−0.008), meaning it generalizes better on test data than training data, which explains its near-zero privacy leakage. In a higher-overfitting scenario (e.g., a deeper GBT), the relationship between generalization gap and MI AUC would be more pronounced.

(c) L₂ regularization tradeoff: Varying C from 0.01 to 10.0 produces MI AUC values of 0.493–0.495 — all effectively random. L2 regularization has no meaningful effect here because LR already doesn't overfit. Test AUC is also stable (0.732–0.735), and AIR stays between 1.946–1.966, meaning regularization introduces no fairness side-effects in this case. The practical takeaway is that for models that already generalize well (like this LR), L2 regularization is a low-cost, zero-risk control. For models with larger generalization gaps, it would be expected to reduce MI AUC more noticeably.

Part 4 — Reflection

Highest-risk finding: The label-flip poisoning attack on African-American defendants is the most dangerous finding. At 30% poison rate, AIR rises from 1.961 to 3.010 while AUC drops only 0.34pp. PSI on all 7 features = 0.0 across all poison rates. This means the attack is simultaneously: (1) undetectable by AUC monitoring, (2) undetectable by PSI drift monitoring, and (3) practically achievable — flipping 345 out of 1,151 eligible training labels. The combination of high impact, low detectability, and low attacker effort makes this the most operationally dangerous finding of the three parts.

Proactive mitigation — Label Auditing + Distribution Checks: Before each model retrain, check the label distribution per racial group for anomalies. From Part 2: flipping 345 AA high-risk labels (30%) raises AIR by +1.05. Even at 5% (57 labels), AIR rises by 0.05 with no AUC signal. A label audit that flags when the high-risk rate for any group drops more than 5% relative to the previous training cycle would catch this attack before the model is retrained. L2 regularization was also tested (Part 3c) but showed no effect here since LR already has minimal overfitting.

Reactive mitigation — AIR Change Monitoring: An AIR dashboard that alerts on changes greater than 0.20 between consecutive model versions would catch the poisoning attack at the first meaningful intervention. From Part 2: the AA-target attack increases AIR by +1.05 at 30% poison rate — well above a 0.20 change threshold. Monitoring absolute AIR change rather than absolute AIR level avoids false-alarming on the pre-existing baseline disparity (AIR = 1.961).

Disparate impact of mitigations:

- Label auditing may introduce differential scrutiny of African-American records if the anomaly threshold is miscalibrated — auditors would be examining AA label rates more frequently given the group's larger representation in the training set. Clear documentation of audit criteria applied equally across all groups is necessary.
- AIR change monitoring is race-group-specific by design, which is appropriate here. However, setting a single change threshold without adjusting for group sample size could result in false alarms for smaller groups (Hispanic, Asian, Native American) where random variation in label rates is higher.